(Crossley §7.1) simplicial complexes

a 0-simplex is a point
a 1-simplex is a (closed, nondegenerate) line segment
a 2-simplex is a triangle
a 3-simplex is a tetrahedron [draw]

Df given a set S of $k + 1$ vectors $v_0, v_1, \ldots, v_k$ in $\mathbb{R}^N$,

we say S is <u>in general position</u> iff $v_1 - v_0, \ldots, v_k - v_0$ are linearly indep

in this case, the convex hull of S is called the associated <u>k-simplex</u> $\Delta(S)$

[what is the convex hull of $v_0, v_1, \ldots, v_k$?]

$$\{t_0v_0 + t_1v_1 + \ldots + t_kv_k \mid t_0, \ldots, t_k \text{ in } [0, 1] \text{ s.t. } \textstyle\sum_i t_i = 1\}$$

Df given a k-simplex $\Delta(S)$:

a <u>subsimplex</u> of $\Delta(S)$ is a simplex of the form

$\Delta(S')$ for some S' sub S

if $|S'| = k - 1$, then it is called a <u>face</u> of $\Delta(S)$

Rem some books say “face” to mean what we call “subsimplex”

Df boundary of Δ = union of its faces;
interior of Δ = complement of boundary

Df a simplicial complex is a finite collection of simplices $\{\Delta(S)\}_S$ s.t.

1) every point of $\bigcup_S \Delta(S)$ belongs to the interior of $\Delta(S)$ for exactly one S

2) if $\Delta(S)$ is in the collection, then so is every face of $\Delta(S)$

we say that $\bigcup_S \Delta(S)$ is the geometric realization of the simplicial complex

Ex simplicial circle [draw]

three 0-simplices
three 1-simplices
[can we use any fewer?]

Ex simplicial square [draw]

four 0-simplices
five 1-simplices
two 2-simplices

Ex simplicial annulus [draw]

six 0-simplices
twelve 1-simplices
six 2-simplices

Df a triangulation of a space X is a homeo from K to X, where K is the geometric realization of some simplicial complex

Rem an n-manifold is a second-countable, Hausdorff space X s.t. every x in X has an open nbd homeo to R^n

(Radó ’25) 2-folds have triangulations

(Moise ’52) 3-folds have triangulations

(Casson ’90) some 4-folds don’t

(Manolescu ’13) some n-folds don’t, for each $n \geq 5$

(Crossley §7.2)

Df the Euler characteristic/number of a simplicial complex is

$$e_0 - e_1 + e_2 - e_3 + \ldots$$

where e_k = # of k-simplices in the collection

Ex simplicial circle: $\chi = 0$

Ex simplicial square: $\chi = 1$

Ex simplicial annulus: $\chi = 0$

Thm (Poincaré) the Euler characteristic only depends on the geometric realization K, not the collection of simplices

so we sometimes denote it by $\chi(K)$

Prob show by example that by contrast, $e_0, e_1, \ldots$ do depend on the collection

motivation: Euler's formula $V - E + F = 2$

Ex simplicial sphere: $\chi = 2$

- four 0-simplices
- six 1-simplices
- four 2-simplices

Warning vertices are always 0-simplices
edges are always 1-simplices
but faces need not be 2-simplices

however, can always triangulate each face

Ex [Example 7.8 in Crossley]
simplicial torus: $\chi = 0$

- nine 0-simplices
- twenty-seven 1-simplices
- eighteen 2-simplices